THOMAS TRAN

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EDUCATION

University of Michigan College of Engineering **Ann Arbor, MI**
Bachelor's of Science in Computer and Electrical Engineering *Expected May 2026*

- **Selected Coursework:** Advanced Embedded Systems, Embedded Control Systems

EXPERIENCE

Symbiotic Engineering and Analysis Lab **Ann Arbor, MI**
Embedded Systems & Hardware Research Engineer *Jan 2026 – Present*

- Programmed a hardware-in-the-loop test bench in C to validate Blue Robotics ESCs and brushless motors, generating 400Hz PWM signals to characterize thrust response across the full RPM range
- Implemented an ESC arming and calibration sequence using timed 1500us PWM pulses, enabling repeatable bench testing of prototype marine propulsion hardware without manual intervention

University of Michigan - CAEN **Ann Arbor, MI**
Classroom Hardware Technician *Aug 2025 – Present*

- Assembled and brought up 10+ custom proprietary PoE camera indicator PCBs, performing through-hole soldering, continuity verification, and firmware flashing for classroom recording status displays
- Diagnosed and resolved 5+ hardware and network faults weekly across 200+ AV-over-IP systems spanning 70+ classrooms, troubleshooting Crestron audio systems, HDMI-over-IP encoders, and control system firmware

Michigan Aeronautical Science Association (MASA) **Ann Arbor, MI**
Avionics Sub-team Radio Lead *Sept 2024 – Feb 2026*

- Directed a 3-person team to develop a bare-metal C driver for the Semtech SX1280 transceiver, implementing register-level radio configuration, FIFO data buffering, and interrupt-driven packet handling
- Traced SX1280 transceiver overheating above 10 dBm to 30 Ω RF traces on a 50 Ω design, validating with Altium stack-up analysis and VNA measurements, and specified a transmit-path power amplifier to compensate for losses
- Performed a 2.4 GHz link budget analysis for a 20 km LoRa telemetry link, finding a -9.6 dB worst-case margin deficit and calculating +12 dB of recovery by specifying a receive-path LNA and reconfiguring modulation to SF12/200kHz
- Developed a Python ground station application with real-time RSSI/SNR monitoring, radio configuration mismatch detection, and CSV telemetry logging over serial to validate SX1280 link performance

PROJECTS

Autonomous Parafoil Guidance & Control System | *Altium Designer, FreeRTOS, STM32*

- Led the hardware design of a 4-layer flight computer PCB in Altium Designer, using a Ground-Signal-Power-Ground stack-up and a star-routed power topology on 2oz copper to isolate 3.3V logic from mechatronic servo stall currents
- Integrated an RF bias-tee circuit and 50 Ω impedance-matched traces into the PCB layout, providing clean DC power to an active NEO-M9N GPS antenna to maximize satellite reception during descent
- Architected a prioritized FreeRTOS system on an STM32H7, writing DMA-driven UART parsers for UBX packets and a custom SPI SHTP driver to execute 100Hz IMU sensor fusion without CPU blocking
- Attained a landing accuracy of <5 meters from a 75m drop by optimizing a 7-state Extended Kalman Filter (EKF) with a sequential update algorithm to reduce matrix computational complexity

IoT Room Occupancy Monitoring System | *Altium Designer, C, ESP32*

- Engineered a 2-layer custom PCB in Altium Designer, integrating an ESP32 microcontroller and a time-of-flight sensor, implementing ground polygon pours, stitching vias, and an RF keep-out zone to ensure reliable Wi-Fi telemetry
- Debugged VL53L1X sensor dead zones during bi-directional and side-by-side foot traffic using diagnostic breakout headers and a 3-LED status circuit for real-time system state validation
- Developed a battery-operated IoT architecture capable of accurately tracking up to 100 individuals in a given space, optimizing component selection to maintain a scalable sub-$100 Bill of Materials (BOM)

SKILLS

Languages: C, C++, Python
Hardware & Platforms: PCB Design (Altium Designer, KiCad), STM32, NXP S32K144, ESP32, FreeRTOS
Software & Tools: MATLAB/Simulink, SPICE, NXP S32 Design Studio, Git